EXPERIMENT 10: PERFORMANCE OPTIMIZATION OF THE STUDENT ATTENDANCE AND ACADEMIC PERFORMANCE TRACKER

AIM

To optimize the performance of the Student Attendance and Academic Performance Tracker in Zoho Creator by improving form loading speed, reducing unnecessary database operations, optimizing Deluge scripts, and ensuring efficient report generation.

PROCEDURE

1. Log in to Zoho Creator and open the Student Attendance and Academic Performance Tracker application.
2. Review the Student Details, Attendance Records, and Academic Performance forms and remove unnecessary fields, scripts, and workflows.
3. Optimize Deluge scripts by avoiding repeated database queries and unnecessary loops.
4. Use criteria-based record fetching to retrieve only the required attendance and academic records.
5. Optimize reports by displaying only essential columns and applying suitable filters.
6. Reduce unnecessary workflow executions and avoid duplicate email notifications.
7. Test the application after optimization to compare loading speed and responsiveness.
8. Verify that all student management, attendance, marks, and reporting features work correctly after optimization.

OPTIMIZATION ACTIONS

Area	Before Optimization	After Optimization
Forms	Duplicate and unused fields	Only required fields are retained and grouped into sections
Deluge Fetch	All attendance records fetched and checked in a loop	Criteria-based fetching retrieves only required records
Reports	All columns displayed without filters	Essential columns and suitable filters are used

Workflows	Notifications triggered on every edit	Notifications triggered only when relevant conditions are met
Dashboard	Unnecessary reports and charts	Only useful reports and charts are displayed

DELUGE OPTIMIZATION EXAMPLE

BEFORE OPTIMIZATION

The following example checks every attendance record to identify students who are absent.

```
for each r in Attendance_Records
{
    if(r.Attendance_Status == "Absent")
    {
        info r.Student;
    }
}
```

Problem: All attendance records are traversed, even when only absent records are required.

AFTER OPTIMIZATION

```
absentRecords = Attendance_Records[Attendance_Status == "Absent"];


for each r in absentRecords
{
    info r.Student;
}
```

Improvement: Criteria-based fetching retrieves only records matching the specified condition, reducing unnecessary processing.

Note: Use the actual form and field link names from your Zoho Creator application.

WORKFLOW OPTIMIZATION EXAMPLE

Before Optimization

Email notifications may be triggered whenever an attendance record is edited, even if the attendance status has not changed.

After Optimization

Configure the workflow to run only when the attendance status changes to Absent, and send a notification only when the relevant condition is satisfied.

This reduces unnecessary notifications and prevents repeated alerts during unrelated edits.

PROCESS DIAGRAM

Figure 10.1: Performance Optimization Workflow

Existing Application
↓
Identify Unnecessary Fields and Operations
↓
Optimize Forms and Deluge Scripts
↓
Apply Criteria-Based Record Fetching
↓
Optimize Reports and Workflows
↓
Test Application Performance
↓
Verify Correct Functionality

BEFORE AND AFTER COMPARISON

Area	Before	After	Benefit
Forms	Duplicate fields	Required fields only	Easier data entry
Deluge Scripts	Process all records	Fetch matching records	Less unnecessary processing
Reports	All columns displayed	Essential columns and filters	Better report usability
Workflows	Repeated notifications	Condition-based notifications	Fewer duplicate alerts
Dashboard	Unnecessary components	Relevant charts and reports	Improved usability

VERIFICATION CHECKLIST

Test	Expected Result
Submit student details	Student record is saved correctly

Add attendance record	Attendance record is saved successfully
Update academic marks	Updated marks appear in the report
Open absent students report	Only absent records are displayed
Open low-attendance report	Students below the configured threshold are displayed
Edit an unrelated field	No unnecessary notification is triggered
Open dashboard	Required reports and charts display correctly

RESULT

The Student Attendance and Academic Performance Tracker was optimized by removing unnecessary fields, using criteria-based record fetching, reducing report columns, and minimizing unnecessary workflow executions. The application was tested to verify that student management, attendance tracking, academic reporting, and dashboard features continued to function correctly.

CONCLUSION

Performance optimization in Zoho Creator improves application maintainability and reduces unnecessary processing. By optimizing forms, Deluge scripts, reports, and workflows, the Student Attendance and Academic Performance Tracker can provide a more efficient and responsive experience for administrators and faculty.